

5.2

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
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In [2]: def f(x):
return(x**4-2*x+1)
```

```
In [3]: def trap(func, a, b, steps):
dx = (b-a)/steps
x = np.linspace(a, b, steps+1)
y = func(x)
s = dx/2*(func(a)+func(b))
s = s + np.sum(y[1:steps]*dx)
return(s)

def simpl(func, a, b, steps):
if steps%2!=0:
steps = steps + 1
print('i added one step')
dx = (b-a)/steps
x = np.linspace(a, b, steps+1)
y = func(x)
s = dx/3*(func(a)+func(b))
s = s + dx/3*(4*np.sum(y[1:steps:2]) + 2*np.sum(y[2:steps-1:2]))
return(s)
```

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In [4]: trap(f, 0, 2, 10)
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Out[4]: 4.506560000000001
```

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In [5]: simpl(f, 0, 2, 10)
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Out[5]: 4.400426666666667
```

fractional error

$$\% \text{ error} = \frac{\text{true value} - \text{estimate}}{\text{true value}} \times 100$$

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In [25]: value = abs(simpl(f, 0, 2, 10) - 4.4)/4.4*100
print(f'the percent error is {value:.4f}%')
```

the percent error is 0.0097%

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In [26]: value2 = abs(simpl(f, 0, 2, 100) - 4.4)/4.4*100
print(f'the percent error is {value2:.4f}%')
```

the percent error is 0.0000%

In []: